

Density-Aware Geodesic Distance Literature Map

Density-Aware Geodesic Distance Literature Map

Executive Map

Historical Spine

Construct 1: Continuous Density-Weighted Path Metrics

Construct 2: Fermat Distances and PWSPD

Construct 3: Density-Based Persistent Homology

Construct 4: Graph Geodesics, Neighbor Selection, and Local Scaling

Isomap-style graph geodesics

kNN, mKNN, iKNN, and adaptive k

Local scaling and density-aware kernels

Mutual reachability distances

Construct 5: Diffusion Maps and Diffusion Distance

Construct 6: PHATE Potential Distance

Construct 7: Random-Walk, Commute, Resistance, Heat-Kernel, and PPR Distances

Commute time and resistance distance

Heat-kernel distances

Personalized PageRank and random-walk-with-restart distances

Construct 8: Density-Aware Oracle Truth for Synthetic Benchmarks

Practical Adoption Review

Microbiome, 16S rRNA, Metagenomics, and Compositional Counts

Single-Cell Genomics and Developmental Trajectories

Spatial Transcriptomics

Ecology and Community Composition

Medical Imaging, Morphology, and Biological Shape

Non-Biological Applied Domains

Relevance to gflow Design

Synthesis Table

Source Index

Density-Aware Geodesic Distance Literature Map

Date: 2026-05-04

This memo is Deliverable 1 for the density-aware geodesic-distance research track. It maps density-aware and density-adjacent distance constructions, then reviews practical adoption in biology-facing data analysis. The main conclusion is conservative: density-

aware geodesic distances are not new, but explicit Fermat/PWSPD-style geodesic distances appear underused in microbiome and metagenomic workflows compared with diffusion, PHATE, UMAP, Aitchison/UniFrac, and other standard beta-diversity approaches.

Executive Map

The closest mathematical match to the current gflow proposal

$$w_{ij} = \frac{\ell_{ij}}{\rho_{ij}^\alpha}.$$

is the continuum Fermat / power-weighted shortest-path metric

$$d_{\rho,\alpha}(x, y) = \inf_{\gamma: x \rightarrow y} \int_{\gamma} \rho(\gamma(s))^{-\alpha} ds.$$

If the data are sampled from an intrinsic d -dimensional manifold with density ρ , and one uses the unrooted Fermat convention

$$d_{F,p}(x, y) = \inf_{\gamma: x \rightarrow y} \int_{\gamma} \rho(\gamma(s))^{-(p-1)/d} ds,$$

then

$$\alpha = \frac{p-1}{d}, \quad p = 1 + \alpha d.$$

Equivalently, this is the geodesic distance of the conformally changed Riemannian metric tensor

$$g_p = \rho^{-2(p-1)/d} g.$$

Some papers write the tensor exponent as $2(p-1)/d$; the path-length integrand exponent, which corresponds to ℓ / ρ^α , is half of that.

There is a convention trap. Some Fermat/PWSPD papers define the discrete distance as

$$\ell_p(x, y) = \min_{\text{path}} \left(\sum_i \|x_i - x_{i+1}\|^p \right)^{1/p},$$

while persistent-homology papers often use the unrooted version

$$d_{X,p}(x, y) = \min_{\text{path}} \sum_i \|x_i - x_{i+1}\|^p.$$

The continuum density-weighted geodesic that matches gflow edge weighting is the unrooted path metric, or ℓ_p under the rooted convention. This should be stated explicitly in any future design memo.

Historical Spine

1. Classical geodesic manifold learning starts from local Euclidean graphs: Isomap estimates manifold geodesic distances by shortest paths on a kNN or epsilon graph before MDS ([Tenenbaum et al. 2000](#)).
2. Density-based distances entered semi-supervised learning as a way to make points close when they are connected through high-density unlabeled data. Sajama and Orlitsky's ICML 2005 work is an early explicit source for density-based Riemannian path metrics ([DBLP record](#)). Bijral, Ratliff, and Srebro later proposed graph-based density-based distances for semi-supervised learning ([arXiv:1202.3702](#)).
3. Fermat distances / power-weighted shortest paths gave a density-sensitive graph-distance estimator that avoids explicit density estimation by raising edge lengths to a power. The theory connects to nonhomogeneous Euclidean first-passage percolation and Fermat's principle in optics ([Groisman et al.](#)).
4. Hwang, Damelin, and Hero analyzed shortest paths through random points and proved convergence of power-weighted sample paths to a density-weighted continuum distance ([arXiv:1202.0045](#)). This is a key bridge from density-based distances to the modern Fermat literature.
5. Recent theory made the continuum limit and spectral behavior precise: Garcia Trillos, Little, McKenzie, and Murphy prove convergence and spectral results for Fermat distances ([JMLR 2024](#)). Fernandez, Borghini, Mindlin, and Groisman use Fermat distance for intrinsic persistent homology ([JMLR 2023](#)). Little, McKenzie, and Murphy analyze PWSPDs and kNN graph spanners ([SIAM MDS 2022](#), [arXiv](#)).
6. Diffusion geometry follows a parallel lineage. Diffusion maps define a Markov operator on a kernel graph and compare random-walk transition distributions; the alpha normalization controls whether sampling density is exploited or removed ([Coifman and Lafon 2006](#)).
7. PHATE uses diffusion probabilities and then transforms them into potentials before embedding. It is support-aware and density-adjacent, but it is not a shortest-path geodesic metric ([Moon et al. 2019](#), [PHATE docs](#)).

Construct 1: Continuous Density-Weighted Path Metrics

Objects:

(M, g) : smooth connected Riemannian manifold,
 $\rho : M \rightarrow \mathbb{R}_+$ sampling/support density,
 $\alpha \geq 0$ density penalty exponent,
 γ : piecewise C^1 curve from x to y ,
 ds_g : Riemannian arclength.

Definition:

$$L_{\rho,\alpha}(\gamma) = \int_0^1 \rho(\gamma(t))^{-\alpha} \sqrt{g_{\gamma(t)}(\gamma'(t), \gamma'(t))} dt,$$

$$d_{\rho,\alpha}(x, y) = \inf_{\gamma: x \rightarrow y} L_{\rho,\alpha}(\gamma).$$

If ρ is positive and finite on each connected component, this is a true path metric on that component. It is the Riemannian geodesic distance for the conformal tensor

$$g_\alpha = \rho^{-2\alpha} g.$$

If ρ reaches zero and no floor is imposed, distances crossing zero-density regions can become infinite. If ρ is estimated and clipped by a floor, the result is a metric for the clipped density but may encode the floor more than the biology in tails.

Density role: high-density regions reduce path cost. Low-density regions act as barriers. This is an exploitation of density as support or plausibility, not a normalization that removes density.

Parameter correspondence:

Fermat/PWSPD path integrand : $\rho^{-(p-1)/d} ds,$

$$\text{gflow } \alpha : \alpha = \frac{p-1}{d},$$

$$\text{Riemannian tensor exponent : } 2\alpha = \frac{2(p-1)}{d}.$$

Finite-data construction:

1. Estimate ρ_i or ρ_{ij} in an appropriate space.
2. Build a graph, usually kNN, mutual kNN, iKNN/mKNN, radius, MST/CMST, or a repaired local graph.
3. Define edge weights such as

$$w_{ij} = \frac{\ell_{ij}}{\phi(\rho_i, \rho_j, \rho_{ij})^\alpha}.$$

4. Compute Dijkstra or all-pairs shortest paths.

Strengths: clear mathematical object; direct control of density sensitivity; compatible with edge/path attribution; naturally matches noisy-tube oracle truths where support is part of the geometry.

Failure modes: density-estimation instability; dimensional scaling errors; tail domination for α near 1 in low-density regions; conflating true biological rarity with sampling failure; bridges can be made too long or too short depending on density summary.

Diagnostic requirements: density distribution, floor/cap fraction, edge ℓ_{ij} versus w_{ij} , path examples, α sensitivity, estimator sensitivity, tail/outlier influence, and comparison to Euclidean, latent, Fermat, and diffusion distances when available.

Construct 2: Fermat Distances and PWSPD

Discrete rooted PWSPD:

$$\ell_p(x, y; X) = \min_{x=x_0, \dots, x_m=y} \left(\sum_{r=0}^{m-1} \|x_{r+1} - x_r\|^p \right)^{1/p}, \quad p \geq 1.$$

Unrooted Fermat distance:

$$d_{X,p}(x, y) = \min_{\text{path}} \sum_{r=0}^{m-1} \|x_{r+1} - x_r\|^p.$$

For $p = 1$, the complete-graph rooted distance collapses to Euclidean distance; on a restricted local graph, it is the usual Isomap-style graph geodesic. For $p > 1$, many short hops are favored, so paths tend to stay in dense regions. As $p \rightarrow \infty$, the optimization approaches a bottleneck / longest-edge path distance.

Continuum limit:

$$d_{F,p}(x, y) = \inf_{\gamma: x \rightarrow y} \int_{\gamma} \rho(\gamma(s))^{-(p-1)/d} ds.$$

With appropriate normalization, sample Fermat distances converge to the population density-weighted geodesic on an intrinsic d -dimensional manifold. The normalization often involves $n^{-(p-1)/d}$ for the unrooted convention or $n^{-(p-1)/(p d)}$ for the rooted convention, up to a percolation constant depending on p and d .

Hwang, Damelin, and Hero's earlier notation makes the same correspondence explicit:

$$n^{(p-1)/d} d_{X_n,p}(x, y) \rightarrow C(d, p) d_{F,p}(x, y),$$

$$g_p = \rho^{2(1-p)/d} g.$$

The sign in the tensor exponent is the same as $\rho^{-(2(p-1)/d)}$.

Asymptotic and finite-sample behavior:

- JMLR 2024 establishes local convergence of discrete Fermat distances to continuum Fermat distances and spectral convergence of Fermat graph Laplacians ([Garcia Trillos et al.](#)).
- SIAM MDS 2022 proves that PWSPD on a kNN graph can match the complete-graph PWSPD with high probability under k of order $\log n$, with constants depending on intrinsic dimension, density bounds, p , and manifold geometry ([Little et al.](#)).
- Parameter choice is not benign: Chazal et al. emphasize a tradeoff between large enough α/p to follow geometry and small enough values to avoid noise-amplified distance estimation ([arXiv:2311.18663](#)).

Practical strengths:

- No explicit density estimate is required.
- kNN graph computation is feasible for large n .
- Directly implements the geometry-density interpolation wanted by `gflow`.

Practical failures:

- Scaling across n , p , and d is easy to mishandle.
- Large p can become bottleneck-like and unstable.
- Complete-graph intuition and kNN-graph behavior differ if k is too small or if local neighborhoods are distorted by high-dimensional noise.

Relevance to `gflow`: this family should be a named comparator and possibly a special case of density-aware graph distance. The noisy-tube oracle can be parameterized to reproduce its continuum limit by setting $\alpha = (p - 1) / d$.

Construct 3: Density-Based Persistent Homology

Motivation: persistent homology of a point cloud in ambient Euclidean distance is sensitive to embedding distortion and outliers. Fermat distances provide an intrinsic metric that accounts for both manifold geometry and sampling density.

Definition: compute Vietoris-Rips persistence on $(X_n, d_{\{X_n, p\}})$, where $d_{\{X_n, p\}}$ is the sample Fermat distance. In the continuum limit, this approximates persistent homology of $(M, d_{\{F, p\}})$.

Metric type: metric-space input for persistent homology; the Fermat component is a density-weighted geodesic sampling-limit object.

Key source: Fernandez et al. prove convergence of persistence diagrams and robustness advantages for positive-degree homology, then apply the method to time-series pattern recognition and anomaly detection, including ECG signals and canary song ([JMLR 2023](#)).

Adoption status: strong as a theory-plus-application paper for time series; not yet a common microbiome or metagenomics tool.

Relevance to `gflow`: provides a topology-facing benchmark for density-aware distances. For noisy circles, annuli, figure-eights, and gaps, persistence under Euclidean, latent, Fermat, and tube-oracle distances could diagnose whether density weighting preserves or destroys intended topology.

Construct 4: Graph Geodesics, Neighbor Selection, and Local Scaling

Isomap-style graph geodesics

Definition:

$$d_G(i, j) = \min_{\text{paths } i \rightarrow j} \sum_{(u, v) \in \text{path}} \|x_u - x_v\|.$$

This is a graph metric on each connected component. Density enters implicitly through graph topology: dense regions have many short edges, sparse regions have fewer or longer edges. It is not explicitly density weighted.

Important caution: unweighted shortest path distances on random kNN graphs can prefer low-density regions because a fixed number of hops travels farther where points are sparse. This is a useful warning for any gflow graph that replaces length weights with hop counts or over-normalizes edge lengths ([Alamgir and von Luxburg 2012](#)).

kNN, mKNN, iKNN, and adaptive k

kNN graphs are computationally attractive but may over-connect dense regions and under-connect sparse ones. Mutual kNN is more conservative and can fragment. iKNN can repair connectivity but can become too dense. Adaptive k and local repair should be evaluated as graph-topology choices before applying density-weighted edge lengths.

Metric type: graph distance after a neighbor-selection rule.

Density role: implicit through neighbor radii and degree; explicit only if k or edge admission depends on density estimates.

Relevance to gflow: density-aware weighting should not be the only lever. Graph topology decides whether unsupported shortcuts exist at all. Weighting a bad shortcut can help only if the density summary recognizes the unsupported gap.

Local scaling and density-aware kernels

Self-tuning spectral clustering uses a local scale, often the distance to the kth neighbor:

$$A_{ij} = \exp \left(-\frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j} \right).$$

This adapts to multi-scale data without explicit density estimation ([Zelnik-Manor and Perona 2004](#)). In biology, Spectrum develops an adaptive density-aware kernel for clustering single and multi-omic data, using common-neighbor information to strengthen within-density connections and providing R software ([Spectrum, Bioinformatics](#)).

Metric type: kernel/spectral geometry, not a shortest-path geodesic unless converted into graph distances.

Relevance to gflow: local scales are natural density surrogates for edge admission and for robust floors/caps.

Mutual reachability distances

HDBSCAN transforms pairwise distances by local core distances:

$$d_{\text{mreach}}(i, j) = \max\{\text{core}_k(i), \text{core}_k(j), d(i, j)\}.$$

This spreads sparse points apart and stabilizes density-based clustering ([HDBSCAN docs](#)).

Metric type: graph/clustering dissimilarity; commonly used through an MST and cluster hierarchy.

Relevance to gflow: mutual reachability is a robust alternative to ℓ_1 / ρ^α for penalizing low local support. It is especially relevant to bridge admission and outlier handling.

Construct 5: Diffusion Maps and Diffusion Distance

Objects:

$$\begin{aligned} K_\varepsilon(i, j) &= \exp\left(-\frac{\|x_i - x_j\|^2}{\varepsilon}\right), \\ q_i &= \sum_j K_\varepsilon(i, j), \\ K_a(i, j) &= \frac{K_\varepsilon(i, j)}{q_i^a q_j^a}, \\ d_i &= \sum_j K_a(i, j), \\ P_a(i, j) &= \frac{K_a(i, j)}{d_i}. \end{aligned}$$

Here a is the diffusion-map alpha normalization, not the gflow density exponent. The diffusion distance at time t can be written as a weighted distance between transition distributions:

$$D_t^2(i, j) = \sum_k \frac{(P_a^t(i, k) - P_a^t(j, k))^2}{\pi_k}.$$

where π is the stationary distribution or an associated reference measure. Spectrally, this is represented by diffusion coordinates with eigenvalues raised to time t .

Density role:

- $a = 0$: no kernel density renormalization; the limiting operator includes a density drift and random walks are strongly density influenced.
- $a = 1$: approximates the Laplace-Beltrami operator and removes sampling density from the continuum geometry.
- intermediate values interpolate between density-sensitive and density-corrected diffusion. Many implementations default near $a = 0.5$.

pydiffmap describes a as the exponent used in left normalization and supports density functions and variable bandwidths ([pydiffmap docs](#)). datafold explicitly describes $a = 1$ as correcting sampling density ([datafold docs](#)).

Metric type: random-walk / embedding distance, not a shortest-path geodesic. It is a metric or pseudometric depending on retained eigenvectors, time, graph connectivity, and whether identical transition profiles occur.

Practical strengths:

- Integrates many paths rather than choosing one.
- Robust to small local graph noise.

- Very mature in single-cell trajectory work.

Failure modes:

- Diffusion time changes scale and topology perception.
- Density normalization may remove biologically meaningful population mass.
- A random-walk distance can make two points close because they diffuse to the same basin, even if the shortest supported path between them is long.

Relevance to gflow: diffusion distances are essential comparators and possibly useful diagnostics, but should not be labeled geodesic distances unless a heat-kernel small-time/geodesic approximation is explicitly intended.

Construct 6: PHATE Potential Distance

PHATE builds an adaptive affinity graph, constructs a Markov diffusion operator, chooses a diffusion time t , transforms transition probabilities into potentials, and embeds by MDS.

Current software defaults include:

```
knn = 5
decay = 40
n_landmark = 2000
t = "auto"
gamma = 1
n_pca = 100
knn_dist = "euclidean"
```

t is the power of the diffusion operator and controls diffusion scale. $\gamma = 1$ gives log potential; $\gamma = 0$ gives square-root potential ([PHATE docs](#)).

Formal sketch:

$$P_t = P^t.$$

$$U_\gamma(P_t(i, \cdot)) = \begin{cases} \log P_t(i, \cdot), & \gamma = 1, \\ \text{power or square-root-like potential}, & \gamma \approx 0, \end{cases}$$

$$D_{\text{PHATE}}(i, j) = \|U_\gamma(P_t(i, \cdot)) - U_\gamma(P_t(j, \cdot))\|_2.$$

Metric type: potential/embedding distance derived from diffusion probabilities. It is support-aware and density-adjacent, but not a shortest-path geodesic.

Density role: mediated by kNN graph construction, kernel tails, Markov normalization, diffusion time, and potential transform. High-density paths increase transition probabilities, but row normalization and diffusion can also smooth or erase density contrast.

Practical adoption: strong in single-cell, mass cytometry, Hi-C, and some microbiome work. The PHATE paper reports gut microbiome data among applications ([Moon et al. 2019](#)).

Relevance to gflow: PHATE should be exposed as a diffusion/potential comparator, not as a density-weighted geodesic. A future gflow PHATE-like option should name the output as potential or diffusion distance.

Construct 7: Random-Walk, Commute, Resistance, Heat-Kernel, and PPR Distances

Commute time and resistance distance

For a connected weighted graph with Laplacian L , effective resistance is

$$R_{ij} = (e_i - e_j)^T L^+ (e_i - e_j).$$

where L^+ is the Moore-Penrose pseudoinverse. Commute time is proportional to resistance:

$$C_{ij} = \text{vol}(G) R_{ij}.$$

Metric type: graph metric for resistance; random-walk expected-time distance for commute.

Density role: high-conductance, many-path regions have low resistance. This is not a shortest path; parallel paths matter.

Failure mode: in large random neighborhood graphs, commute and hitting times can become dominated by vertex degrees rather than meaningful global geometry ([von Luxburg et al. 2014](#)).

Biology adoption: landscape ecology uses resistance and circuit theory to model movement and gene flow through environmental resistance surfaces; this is explicit environmental resistance, not sample-density weighting. R packages such as [gdistance](#) implement commute/resistance distances ([gdistance docs](#)).

Heat-kernel distances

On a manifold, the heat kernel $H_t(x, y)$ is linked to geodesic distance by Varadhan's small-time formula:

$$d_g(x, y)^2 = -\lim_{t \rightarrow 0} 4t \log H_t(x, y).$$

On graphs, one often uses

$$H_t = \exp(-tL).$$

Distances can be formed from rows of H_t , from heat-kernel signatures $H_t(x, x)$, or from small-time geodesic approximations.

Metric type: diffusion/heat distance or shape descriptor; only in a small-time limit does it recover geodesic distance.

Adoption: heat-kernel signatures are well established for shape analysis and medical/biological morphology, but usually as descriptors for matching, retrieval, segmentation, or classification rather than density-aware sample geodesics ([Sun et al. HKS page](#)).

Personalized PageRank and random-walk-with-restart distances

PPR solves

$$r_i = \beta e_i + (1 - \beta) r_i P.$$

Distances or similarities are built from personalized stationary vectors. They emphasize local graph basins and are useful for ranking, clustering, and semi-supervised propagation.

Heat-kernel PageRank replaces the geometric mixture of walk lengths with a Poisson mixture:

$$h_{t,i} = e_i \exp(-t(I - P)).$$

Metric type: usually asymmetric proximity; symmetric variants may be pseudometrics after transformation.

Biology adoption: random-walk-with-restart and local random walks are common in network biology and single-cell fate tools, but their density role is generally implicit through graph construction.

Relevance to gflow: random-walk clouds from a vertex can be used as local volume/density probes, but that should be treated as a diagnostic or density estimator unless a distance is explicitly defined and tested.

Construct 8: Density-Aware Oracle Truth for Synthetic Benchmarks

For a known generative density on a manifold, tube, grid, or graph, define

$$d_{\text{oracle}}(x, y) = \inf_{\gamma: x \rightarrow y} \int_{\gamma} \rho(\gamma(s))^{-\alpha} ds,$$

or build a dense oracle graph with edge weights

$$w_{ab} = \frac{\|z_a - z_b\|}{\rho(\text{midpoint}(a, b))^\alpha}.$$

Metric type: synthetic truth metric, not a data estimator.

Conditions:

- positive density floor or explicit support boundary;
- convergence checks over oracle grid resolution;
- transparent sample-to-oracle attachment;
- sensitivity over α , density floor, cap, and attachment rule.

Relevance to gflow: the noisy-tube oracle is a correct and useful internal object if presented as an oracle benchmark for a chosen density-weighted truth, not as a new distance theory.

Practical Adoption Review

Microbiome, 16S rRNA, Metagenomics, and Compositional Counts

Observed adoption:

- Standard microbiome workflows overwhelmingly use beta-diversity distances: Bray-Curtis, Jaccard, Jensen-Shannon, weighted/unweighted UniFrac, generalized UniFrac, Aitchison/robust Aitchison, and related ordinations. These are not sample-density geodesics.
- PHATE has concrete microbiome use. A large gut microbiome analysis used PHATE to map local ecological states into global branches and then performed pseudotime analysis along those branches ([Nature Communications 2023](#)). The PHATE effect is implicit diffusion/potential geometry, not an explicit $\rho^{-\alpha}$ geodesic.
- tmap uses topological data analysis and network representation for population-scale microbiome stratification ([Genome Biology 2019](#)). It is important adoption context for graph/TDA microbiome workflows, but not a Fermat geodesic method.
- Diffusion maps appear in ecological/microbiome-adjacent trait-space work, including bacterial metabolic niche maps based on genome-scale metabolic traits combined with Earth Microbiome Project 16S community data ([Fahimipour and Gross 2020](#)).
- Compositional methods such as DEICODE / robust Aitchison PCA address sparse microbiome counts directly, but are not graph geodesic density metrics ([Martino et al. 2019](#)).

Evidence gap:

- I found little evidence that Fermat distance, PWSPD, or explicit $\ell_1 / \rho^{-\alpha}$ graph geodesics are standard in microbiome, 16S, metagenomic, or metatranscriptomic analysis.
- This gap is a plausible gflow opportunity, but only if density estimation is compositional-data-aware and benchmarked against Aitchison/UniFrac/PHATE.

Scientific questions motivating use:

- gradients and branches of community states;
- stratification of host/environment associations;
- visualization and ordination;
- pseudotime-like ecological transitions.

Software/defaults/diagnostics:

- PHATE and phateR provide defaults and usable software.
- Microbiome-specific density-aware geodesic diagnostics appear largely absent.

Single-Cell Genomics and Developmental Trajectories

Observed adoption:

- Diffusion maps and diffusion pseudotime are deeply adopted. *destiny* computes cell-to-cell transition probabilities, applies density normalization, and uses eigendecomposition for diffusion components ([destiny paper](#), [vignette](#)).
- DPT reconstructs lineage branching from diffusion geometry ([Haghverdi et al. 2016](#)).
- PHATE is heavily adopted for visualizing high-dimensional biological transitions, including scRNA-seq and mass cytometry ([Moon et al. 2019](#)).
- MAGIC uses graph diffusion for denoising/imputation rather than reporting a distance matrix ([MAGIC paper](#)).
- Palantir, Wishbone, MARGARET, and related tools use diffusion maps, Markov chains, waypoints, random walks, and fate probabilities rather than explicit density-weighted shortest paths.
- scPMP is the clearest biology adoption of PWSPD. It uses power-weighted path metrics for scRNA-seq clustering and visualization, computes shortest paths on kNN graphs with edge weights $||x_i - x_j||^p$, and provides software and defaults ([PLOS Computational Biology 2024](#)).
- DensityPath builds a density landscape after dimensional reduction, extracts high-density representative cell states, and connects them by geodesic MST on the density surface ([Bioinformatics 2019](#)).

Density interpretation:

- Single-cell diffusion methods often correct away sampling density because rare and abundant populations should both be represented.
- scPMP and DensityPath are the strongest precedents for exploiting density as biological support.

Relevance to gflow:

- Single-cell gives both caution and precedent. It shows adoption is possible, but also that the field distinguishes density correction from density exploitation depending on biological question.

Spatial Transcriptomics

Observed adoption:

- Spatial methods commonly use spatial neighbor graphs, graph convolution, diffusion propagation, or physical-distance kernels.
- I found less evidence for Fermat/PWSPD-style density-weighted geodesic distances over spatial transcriptomic cell-state manifolds.

Likely transferable lesson:

- Separate density in measurement space, spatial density of spots/cells, and biological density of states. A single ρ may conflate them.

Ecology and Community Composition

Observed adoption:

- Ecology uses ordination, beta-diversity, distance decay, environmental gradients, and resistance surfaces extensively.
- Diffusion-map dissimilarity has been proposed for ecological community composition when communities have high species turnover and few shared species

(Gault et al. 2023).

- Circuit/resistance distances are common for movement, dispersal, and gene flow through landscapes, but their “density” is environmental conductance or resistance, not sample density.

Relevance to gflow:

- Ecology is conceptually close to microbiome state-space work. It reinforces that support-aware paths are meaningful, but also that the resistance surface must be scientifically interpretable.

Medical Imaging, Morphology, and Biological Shape

Observed adoption:

- Heat-kernel and diffusion descriptors are used for shape matching, segmentation, and morphology because they are intrinsic and multiscale.
- These methods generally use geometry of an observed surface or mesh rather than sampling-density-weighted data geodesics.

Relevance to gflow:

- Heat-kernel diagnostics may be useful for graph stability and multiscale structure, but should be kept separate from shortest-path density-weighted distances.

Non-Biological Applied Domains

Observed adoption:

- Fermat/PWSPD has applications in clustering, semi-supervised learning, image segmentation, and time-series pattern recognition.
- Density-based graph distances are also central to DBSCAN, OPTICS, HDBSCAN, and density-based cluster validation.

Transferable lessons:

- Explicit density-aware path metrics work best when the density is meaningful and stable.
- Large density exponents are powerful but fragile.
- kNN graph spanner results justify computational shortcuts, but constants and preprocessing matter in finite biological data.

Relevance to gflow Design

Known/borrowed:

- ℓ_1 / ρ^α is a standard conformal path metric.
- Fermat/PWSPD supplies a density-free estimator of a related continuum metric.
- Diffusion/PHATE supplies support-aware random-walk comparators.

Potentially distinctive:

- compositional and microbiome-aware density estimation;
- diagnostics for floors, caps, tails, and bridge attribution;
- integration with CMST/iKNN/mKNN graph construction and repair;
- noisy-tube oracle benchmarks;
- downstream gradient-flow and conditional expectation analysis.

Immediate recommendations for later deliverables:

1. Parameterize any explicit density-weighted graph metric by α , and report the implied Fermat $p = 1 + \alpha d_{\text{hat}}$ when an intrinsic dimension estimate is available.
2. Implement Fermat/PWSPD as a comparator, not only explicit density weighting.
3. Keep diffusion/PHATE distances in a separate family named “diffusion” or “potential”, not “geodesic”.
4. Require density floors/caps and tail diagnostics before any density-aware distance is used downstream.
5. Benchmark against microbiome-native comparators: Aitchison/robust Aitchison, Bray-Curtis, UniFrac, PHATE, UMAP/densMAP, and graph-geodesic baselines.

Synthesis Table

Family	Density role	Metric type	Should gflow call it geodesic?	Adopt
Continuous integral $\rho^{-\alpha} ds$	Explicitly exploits density as support	Path/geodesic metric if $\rho > 0$	Yes	Theory, supervised learning
Explicit edge reweighting $\ell_1 / \phi(\rho)^\alpha$	Explicitly exploits estimated density	Graph shortest-path metric	Yes, graph geodesic	Limited evidence
Fermat/PWSPD	Implicitly exploits density through edge powers	Sample graph metric with continuum geodesic limit	Yes, with convention stated	Strong scPMP seq
Density-based persistent homology	Uses Fermat metric for topology	Metric-space/TDA object	The distance yes; PH output no	Time series applications
Isomap/kNN geodesic	Density affects topology implicitly	Graph shortest-path metric	Yes, graph geodesic	Broad
mKNN/iKNN/adaptive k	Density affects topology/degree	Graph construction	Only after weights defined paths	Local graph practice
Local scaling/self-tuning kernels	Usually corrects/adapts	Kernel/spectral geometry	No, unless converted	Spectral methods

Family	Density role	Metric type	Should gflow call it geodesic?	Adopt
	to local scale		to paths	
HDBSCAN mutual reachability	Penalizes low local density/core support	Clustering dissimilarity/MST	Not usually	Broad c
Diffusion maps	Can exploit or correct density via alpha	Random-walk/embedding distance	No	Very sti cell
PHATE potential distance	Implicit support through diffusion probabilities	Potential/embedding distance	No	Strong some m
Commute/resistance	Many high-conductance paths reduce distance	Random-walk/electrical graph metric	No	Ecology biology
Heat-kernel distance/signature	Diffusion over geometry	Heat/diffusion descriptor	Only in small-time limit	Shape/r
Personalized PageRank/RWR	Local basin probability	Proximity, often asymmetric	No	Networ
Density-aware oracle	Known generative density defines truth	Synthetic benchmark metric	Yes, if path metric	Internal benchr

Source Index

- Fermat/PWSPD theory: [Hwang et al. arXiv:1202.0045](#), [JMLR 2024](#), [SIAM MDS 2022](#), [arXiv:2012.09385](#), [arXiv:1810.09398](#).
- Fermat persistent homology: [JMLR 2023](#).
- Fermat parameter selection: [arXiv:2311.18663](#).
- Semi-supervised density-based distances: [arXiv:1202.3702](#), [ICML 2005 DBLP](#).
- Diffusion maps: [Coifman and Lafon DOI](#), [pydiffmap docs](#), [datafold docs](#).
- PHATE: [Nature Biotechnology 2019](#), [PHATE docs](#).
- Single-cell diffusion adoption: [destiny Bioinformatics](#), [DPT Nature Methods](#), [MAGIC Cell](#).
- Single-cell density/path adoption: [scPMP PLOS Computational Biology 2024](#), [DensityPath Bioinformatics 2019](#).
- Microbiome adoption: [Gut microbiome branches with PHATE](#), [tmap Genome Biology 2019](#), [DEICODE/robust Aitchison](#).
- Local scaling and density-aware kernels: [Self-tuning spectral clustering](#), [Spectrum Bioinformatics](#).

- Random-walk and resistance cautions: von Luxburg et al. JMLR, gdistance commuteDistance.